STAFF REVIEW · H446 263 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Copies ACC to mailbox.

2. Stops execution.

3. Without updating a tested state, the loop may not terminate.

4. $1+0+1=10_2$; sum=0, carry-out=1.

I Do — source worked steps

1. Predict first — INP sets ACC to 0

2. Check next state — BRZ ZERO takes branch

3. Check next state — LDA ONE loads 1

4. Check next state — OUT displays 1.

We Do — guided CHECK

1. Predict first
INP makes ACC 4
Check next state
BRZ not taken
OUT prints 4
HLT stops execution.

You Do — source mark schemes

1. BRA unconditional [1]; BRZ if ACC zero [1]; BRP if ACC non-negative [1]; taken branch changes instruction address [1].

2. ACC 3 [1]; BRZ not taken [1]; subtract 1 -> 2 [2]; OUT prints 2 [1]; HLT [1].

3. BRA repeats [1]; no progress/exit [1]; initialise count [1]; decrement [1]; store update [1]; test an exit condition [1].

Exit Ticket — source answer

1. False. It needs a reachable exit condition.